

Scaling Comparisons for the filtering algorithms

Evan Tate Paterson Hughes

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Currently, the Kalman filter, the Information filter, the Square-root filter, and the square-root-information filter are all implemented.

We will use a simulated data set, from cosine basis functions over 100 frequencies and invariant kernel. See figure (?) for the 6 time points of the simulated process.

We then filter this data, and time the implemented filters to compute the marginal log likelihood. We run the algorithms with a sequence of model parameters that are close to the ones used to simulate the data, with added noise to avoid the memoisation the JAX uses.

We simulate $n = 300$ observations per time point in total, and redact observations in order to show the compute times for each n between 50 and 300. These are computed in 64 bit for stability, and the results are in figure (?)

Additionally, to demonstrate the square root filters stability, we also fix n and progressively increase n . The data is simulated from a high-dimensional process basis, and then filtered using models with a lower r , increasing in square between 5^2 and 15^2 . The results are shown in figure (?).

Chapter 1

Laptop RTX 4060 (Lovelace)



```

import pandas as pd
import plotly.express as px

# Load the datasets
df64 = pd.read_csv('data/varying_n_laptop_64.csv')
df32 = pd.read_csv('data/varying_n_laptop_32.csv')

# Select and melt relevant columns from df64
df64_melted = df64.melt(
    id_vars="n",
    value_vars=df64.columns[3:7],
    var_name="Filter",
    value_name="Time"
)
df64_melted["Precision"] = "64-bit"

# Select and melt the two columns from df32
df32_melted = df32.melt(
    id_vars="n",
    value_vars=[df32.columns[4], df32.columns[6]],
    var_name="Filter",
    value_name="Time"
)
df32_melted["Precision"] = "32-bit"

# Combine both
df_combined = pd.concat([df64_melted, df32_melted], ignore_index=True)

rename_map = {
    "kalman_run": "Kalman",
    "sqrt_run": "Square-root",
    "inf_run": "Information",
    "sqinf_run": "Sqrt Information",
}

# Apply renaming
df_combined["Filter"] = df_combined["Filter"].replace(rename_map)

# Plot using Plotly Express
fig = px.line(
    df_combined,
    x="n",
    y="Time",
    color="Filter",
    line_dash="Precision",
    labels={"n": "number of observations", "Time": "Time (seconds)"},
    title="Filters Comparison with Varying n"
)

fig.write_image("figure/scale_test_n_laptop.png")
fig.show()

```

/tmp/ipykernel_11306/2547227509.py:51: DeprecationWarning:

Support for Kaleido versions less than 1.0.0 is deprecated and will be removed after September 2025.
Please upgrade Kaleido to version 1.0.0 or greater (`pip install 'kaleido>=1.0.0'` or `pip install 'plotly`

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Chapter 2

NCC (Ampere)

```

import pandas as pd
import plotly.express as px

# Load the datasets
df64 = pd.read_csv('data/varying_n_ncc_64.csv')
df32 = pd.read_csv('data/varying_n_ncc_32.csv')

# Select and melt relevant columns from df64
df64_melted = df64.melt(
    id_vars="n",
    value_vars=df64.columns[3:7],
    var_name="Filter",
    value_name="Time"
)
df64_melted["Precision"] = "64-bit"

# Select and melt the two columns from df32
df32_melted = df32.melt(
    id_vars="n",
    value_vars=[df32.columns[4], df32.columns[6]],
    var_name="Filter",
    value_name="Time"
)
df32_melted["Precision"] = "32-bit"

# Combine both
df_combined = pd.concat([df64_melted, df32_melted], ignore_index=True)

rename_map = {
    "kalman_run": "Kalman",
    "sqrt_run": "Square-root",
    "inf_run": "Information",
    "sqinf_run": "Sqrt Information",
}

# Apply renaming
df_combined["Filter"] = df_combined["Filter"].replace(rename_map)

# Plot using Plotly Express
fig = px.line(
    df_combined,
    x="n",
    y="Time",
    color="Filter",
    line_dash="Precision",
    labels={"n": "number of observations", "Time": "Time (seconds)"},
    title="Filters Comparison with Varying n"
)

fig.write_image("figure/scale_test_n_ampere.png")
fig.show()

```

/tmp/ipykernel_11306/3675275947.py:51: DeprecationWarning:

Support for Kaleido versions less than 1.0.0 is deprecated and will be removed after September 2025.
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Chapter 3

NCC (Turing)

```

import pandas as pd
import plotly.express as px

# Load the datasets
df64 = pd.read_csv('data/varying_n_turing_64.csv')
df32 = pd.read_csv('data/varying_n_turing_32.csv')

# Select and melt relevant columns from df64
df64_melted = df64.melt(
    id_vars="n",
    value_vars=df64.columns[3:7],
    var_name="Filter",
    value_name="Time"
)
df64_melted["Precision"] = "64-bit"

# Select and melt the two columns from df32
df32_melted = df32.melt(
    id_vars="n",
    value_vars=[df32.columns[4], df32.columns[6]],
    var_name="Filter",
    value_name="Time"
)
df32_melted["Precision"] = "32-bit"

# Combine both
df_combined = pd.concat([df64_melted, df32_melted], ignore_index=True)

rename_map = {
    "kalman_run": "Kalman",
    "sqrt_run": "Square-root",
    "inf_run": "Information",
    "sqinf_run": "Sqrt Information",
}

# Apply renaming
df_combined["Filter"] = df_combined["Filter"].replace(rename_map)

# Plot using Plotly Express
fig = px.line(
    df_combined,
    x="n",
    y="Time",
    color="Filter",
    line_dash="Precision",
    labels={"n": "number of observations", "Time": "Time (seconds)"},
    title="Filters Comparison with Varying n"
)

fig.write_image("figure/scale_test_n_turing.png")
fig.show()

```

/tmp/ipykernel_11306/3756852843.py:51: DeprecationWarning:

Support for Kaleido versions less than 1.0.0 is deprecated and will be removed after September 2025.
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